Virology 587 (2023) 109860

Contents lists available at ScienceDirect

Virology

journal homepage: www.elsevier.com/locate/virology

H5N1 highly pathogenic avian influenza clade 2.3.4.4b in wild and domestic birds: Introductions into the United States and reassortments, December 2021–April 2022

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ARTICLE INFO

Handling Editor: Dr. Jasmine Tomar

Keywords:
H5N1
Highly pathogenic avian influenza
Emerging infectious diseases
Phylogenetic analysis
Poultry
Wild birds

ABSTRACT

Highly pathogenic avian influenza viruses (HPAIVs) of the A/goose/Guangdong/1/1996 lineage H5 clade 2.3.4.4b continue to have a devastating effect on domestic and wild birds. Full genome sequence analyses using 1369 H5N1 HPAIVs detected in the United States (U.S.) in wild birds, commercial poultry, and backyard flocks from December 2021 to April 2022, showed three phylogenetically distinct H5N1 virus introductions in the U.S. by wild birds. Unreassorted Eurasian genotypes A1 and A2 entered the Northeast Atlantic states, whereas a genetically distinct A3 genotype was detected in Alaska. The A1 genotype spread westward via wild bird migration and reassorted with North American wild bird avian influenza viruses. Reassortments of up to five internal genes generated a total of 21 distinct clusters; of these, six genotypes represented 92% of the HPAIVs examined. By phylodynamic analyses, most detections in domestic birds were shown to be point-source transmissions from wild birds, with limited farm-to-farm spread.

1. Introduction

Avian Influenza viruses (AIVs) are eight segmented negative-sense RNA viruses belonging to the genus *Alphainfluenzavirus* of the family *Orthomyxoviridae*. AIVs can be classified as either low or highly pathogenic (LP or HP) depending on the severity of the disease they cause in chickens (*Gallus gallus domesticus*) but with similar outcomes in other gallinaceous species. Historically, classical HPAIVs have emerged in domestic poultry after a LPAIV was introduced from wild aquatic birds, and HPAIVs are not widely maintained in wild bird populations (Swayne and Suarez, 2000). However, the A/goose/Guangdong/1/96 (Gs/GD) H5 lineage HPAIVs have caused substantial losses in both poultry and wild bird populations for decades. Since the emergence of the virus in 1996, the H5 Gs/GD lineage viruses have been maintained and evolved in domestic poultry, including domestic ducks and geese, initially in Asia, with subsequent spillover into migratory wild birds. The repeated spillover of the H5 Gs/GD lineage viruses from poultry to wild aquatic birds resulted in the persistence of variants in wild birds that gave rise to four intercontinental spread events to European and African countries through wild bird migration (Lee et al., 2021; Sims et al., 2017). The hemagglutinin (HA) gene of this H5 Gs/GD lineage has been assigned a unified nomenclature by the World Health Organization (WHO) to track the evolution of phylogenetically distinct clades (World Health Organization, 2020). The current H5 2.3.4.4 clade has several defined subclades, some of which have undergone substantial reassortment with LPAIVs of wild birds across several flyways.

Intercontinental spread of the H5 clade 2.3.4.4 Eurasian lineage viruses caused the first outbreak in North America in 2014/2015, and in 2016/2017 affected Europe, Africa, and Asia. The distinct clades that caused these long-distance waves have been defined by the WHO unified

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https://doi.org/10.1016/j.virol.2023.109860
Received 8 June 2023; Received in revised form 31 July 2023; Accepted 1 August 2023
Available online 2 August 2023
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nomenclature as clade 2.3.4.4c for the 2014/2015 viruses and clade 2.3.4.4b for the 2016/2017 viruses The 2014/2015 wave introduced a fully Eurasian (EA) H5N8 virus from East Asia to North America likely across the Bering Strait to breeding grounds in Alaska (Lee et al., 2015, 2016). The virus initially caused a small number of poultry and wild bird cases, and reassortments with LPAIV gene segments from North American (AM) wild birds quickly emerged, including the neuraminidase (NA) genes (Lee et al., 2016). In the spring of 2015, a reassortant EA/AM H5N2 clade 2.3.4.4c virus became predominant and spread to poultry affecting >50 million birds in the United States (U.S.) before its eradication from poultry in June 2015 (Lee et al., 2018). The H5 clade 2.3.4.4c virus was only sporadically detected in wild birds after June 2015 with the last detection in December 2016 (USDA-APHIS, 2017). In contrast to the 2014/2015 wave, the H5 clade 2.3.4.4b viruses, which did not reach North America, caused a distinct epidemic wave in wild bird population and poultry in Europe until 2018, and a separate epidemic in East Europe in 2018/2019, with few to no detections between waves before the emergence of H5N1 clade 2.3.4.4b in 2020 (EFSA et al., 2021). Of note, the 2016/2017 wave viruses caused substantial wild bird mortality events compared to the clade 2.3.4.4c viruses (Verhagen et al., 2021), and experimentally, the virus was more pathogenic in several duck species tested (Koethe et al., 2020; Leyson et al., 2021).

In October 2020, a distinct H5N1 2.3.4.4b virus was first detected in wild birds in the Netherlands (Lewis et al., 2021). Since that time, several European countries have reported H5N1 detections in wild birds prior and up to December 2021 (EFSA et al., 2021). Four major reassortant viruses were identified (Lycett et al., 2020) with NA gene reassortants including H5N1, H5N3, H5N4, H5N5, H5N6 and H5N8 (Engelsma et al., 2022; Lewis et al., 2021). The continued circulation of H5 clade 2.3.4.4b viruses was apparently limited to Eurasian flyways until late 2021. Before the first detection in the U.S., H5N1 clade 2.3.4.4b was found in poultry and a coastal seabird in Newfoundland and Labrador, Canada. The high genetic identity of Canadian isolates with a wild bird H5N1 virus from Ireland (A/whooper swan/Ireland/032444 22VIR1325-2/2021) indicated transatlantic spread via wild birds (Caliendo et al., 2022), possibly through Iceland (Gunther et al., 2022). Shortly thereafter, the first H5N1 clade 2.3.4.4b virus introduction in the U.S. was reported in wild waterfowl in South Carolina and North Carolina in samples collected at the end of December 2021 as part of hunter-harvest duck surveillance (Bevins et al., 2022). Wild bird detections continued eventually throughout all 48 states in the continental U.S. and Alaska (USDA-APHIS, 2023a) as well as sporadic spillovers into wild mammals (Elsmo et al., 2023; OFFLU, 2023; USDA-APHIS, 2023b). A large number of domestic birds had been affected as of March 2023, including 321 commercial and 469 backyard flocks, for a total of 58 million birds (USDA-APHIS, 2023a).

Examination of the H5N1 clade 2.3.4.4b virus sequences indicates that these viruses likely emerged from a reassortment with EA wild bird LPAIVs and were first detected in the Netherlands in October 2020 (Engelsma et al., 2022). The virus then dispersed throughout Europe primarily via wild migratory bird movements causing mortality events in wild birds and spilling over into poultry, subsequently diverging into multiple genotypes arising from reassortments with wild bird LPAIVs (EFSA et al., 2021). Separately, an H5N1 clade 2.3.4.4b virus from a moribund bald eagle found in British Columbia, Canada, in February 2022 was found to be genetically related to a virus isolated in Hokkaido, Japan, in January 2022 (Alkie et al., 2022). These two phylogenetically distinct viruses from the same H5 clade 2.3.4.4b represent separate introductions to Canada via the Pacific and Atlantic flyways. This study aimed to examine the genetic characteristics of the H5N1 HPAIVs isolated in the U.S. from December 2021 to April 2022 with the further goal of elucidating its origins and the dynamics of its spread.

2. Materials and methods

2.1. Sample collection and sequencing

From December 2021 to April 2022, over 13,200 wild bird samples were collected from hunter-harvested birds, live captured birds, and bird mortality events by the U.S. Department of Agriculture (USDA) Wildlife Services, state, and academic partners (USDA-APHIS, 2023d) (Supplementary Fig. 1). For poultry, the surveillance program is conducted as described by the National Poultry Improvement Program (https://www.poultryimprovement.org/), which guides testing for commercial poultry sectors and game birds; testing at bird auctions and markets is guided by the Prevention and Control of H5 and H7 Avian Influenza in the Live Bird Marketing System: Uniform Standards for a State-Federal-Industry Cooperative Program (Jagne et al., 2021). During the 2021–2022 H5 2.3.4.4b outbreak, continuity of business is guided the Secure Poultry Supply Plans (USDA-APHIS-VS, 2013), and outbreak response and policy is guided by the HPAI Response Plan: The Red Book (USDA-APHIS-VS, 2017). From January 2022 to March 2022 (3-month timeframe) an estimated 500,000 federally funded tests were conducted in support of routine surveillance for U.S. poultry and in response to the 2022 HPAI outbreak.

Samples were screened by real-time reverse transcription polymerase chain reaction (RT-PCR) targeting conserved region(s) of the influenza A virus (IAV) matrix gene (Spackman, 2014) by members of the National Animal Health Laboratory Network (NAHLN), as well as university and academic partners. Further avian influenza subtyping was performed using at least one IAV H5 subtyping assay: the NAHLN H5 assay targeting H5 viruses from North America, Eurasia, and Mexico (Pedersen et al., 2014; Spackman and Suarez, 2008), and for poultry samples, an H5 2.3.4.4b-specific PCR developed in collaboration with USDA Agriculture Research Service (ARS) Southeast Poultry Research Laboratory (SEPRL; Real-Time RT-PCR Assay for the Detection of Goose/Guangdong lineage Influenza A subtype H5 clade 2.3.4.4; NVSL-WI-1732).

Samples, or in some cases nucleic acid extracts, with non-negative influenza A results were shipped to the to the National Veterinary Services Laboratories (NVSL, Animal and Plant Health Inspection Service, USDA in Ames, Iowa, U.S.) for confirmation and characterization. Testing at NVSL included an H5 clade 2.3.4.4 pathotyping assay and an assay targeting N1 for rapid pathotyping (SEPRL; Real-Time RT-PCR Assay for Pathotyping Goose/Guangdong lineage Influenza A subtype H5 clade 2.3.4.4; NVSL-WI-1767) and NA subtyping (SEPRL; Real-Time RT-PCR Assay for the Detection of Eurasian-lineage Influenza A Subtype N1; NVSL-WI-1768). Influenza A viruses were sequenced directly from samples as previously described (Bevins et al., 2022) and RAxML was used to generate phylogenetic trees. Tables of single nucleotide polymorphisms (SNPs) were created using a vSNP pipeline developed by NVSL (https://github.com/USDA-VS/vSNP). This tool utilizes a curated BLAST database and reference file to assign each segment group and the overall constellation and genotype using a table of established genotypes. A cutoff of 2% nucleotide difference from the closest reference sequence is used to identify new reassorted segments, which are then reviewed using segment-based phylogenetic trees. New references and genotypes were assigned as needed.

2.2. Sequence data

Available quality sequences were included at the time of data selection. Full genomes sequenced in this study were used for the analyses (n = 1315; 856 from wild birds, 334 from commercial poultry, and 125 from backyard flocks). Additional sequences, genetically close to the H5N1 reassortants, were chosen from the Basic Local Alignment Search Tool (BLAST) analyses in the Influenza Research Database (Squires et al., 2012). For each segment, the top 200 hits were combined with a previously established dataset where a representative sample of all

North American lineage wild bird AIVs were reduced using a cut off value of 98% sequence identity using the CD-HIT Suite (Youk et al., 2022). To trace the non-reassorted H5N1 viruses that crossed the Atlantic Ocean via wild waterfowl (Bevins et al., 2022), 43 viruses genetically close to the four early H5N1 sequences from an Eastern Province of Canada (Newfoundland and Labrador) in December 2021 and Jan 2022 (Caliendo et al., 2022) were chosen from BLAST analyses using the Global Initiative for Sharing All Influenza Database (GISAID). H5 clade 2.3.4.4b sequences from wild birds from Alaska in April 2022 (n = 2), and viruses from Europe (n = 3) and Japan (n = 3) were also included from BLAST analyses in GISAID, as well as one sequence from a Western Province of Canada (British Columbia) in February 2022. A total of 1369 viruses were thus considered for further phylogenetic analyses. Strain names and accession numbers used are summarized in Supplementary Table 1. The number of U.S. wild bird H5N1 viruses included in the analyses was tabulated by species order in Supplementary Table 2.

2.3. Genotype classification

The H5N1 viruses with at least one North American wild bird LPAIV gene segment were used to construct the Maximum Likelihood (ML) phylogenetic trees with the BLAST results and the previously established North American lineage wild bird AIV dataset. The RAxML v8 was used under the general time-reversible (GTR) substitution model and the discrete Gamma model of rate heterogeneity with 1000 bootstrap replicates (Stamatakis, 2014). The resulting ML trees were inspected for clock-likeness by plotting a root-to-tip regression in the TempEst v1.5.1. To find the association between genotype distribution and wild bird movement, the color codes were appended to the ML trees to identify the North American wild bird flyway involved as well as the genotype. Distinct monophyletic clusters within each gene segment were described using the following criteria: (1) a group is monophyletic with bootstrap values greater than 70%, (2) genetic identity within a group is greater than 99% for PB2, NP and NS, and greater than 98% for PB1 and PA; and (3) genetic identities between groups are lower than 99% for PB2, NP, and NS, and lower than 98% for PB1 and PA. The threshold values for within-group and between-group genetic identities were optimized by gene segment to effectively illustrate phylogenetic clusters using bootstrap values and geographical differences. This allowed inclusion of all the viruses sequenced in the present study in the downstream analyses. As a result, each cluster in a gene segment was distinctly labeled in accordance with their geographical lineage, denoted as either 'ea' (Eurasian) or 'am' (American). Then, an organized hierarchical system of numeration was developed for these groups, which corresponded directly to their isolation date order within the constructed phylogenetic tree. This ordering was based on the evolutionary relationships and lineage traced in a phylogenetic tree, as illustrated in Supplementary Fig. 2. Then, the genotype is determined by the gene constellation generated by concatenating the clusters of each virus strain by following gene segment order; PB2|PB1|PA|HA|NP|NA|M|NS. For downstream phylodynamic analyses, gene constellations that accounted for less than 2% (30/1315) of all sequence data were classified as minor genotypes and were excluded.

2.4. Datasets for network analyses and Bayesian phylogenetic trees

All available coding regions of sequences were concatenated using only the gene segments from the same genetic groups, and genes segments resulting from reassortment were omitted. All gene segments were concatenated for the A1, A2, B2, B4, and B5 genotypes (A3 only had two representatives and was excluded). For B1 and B3, only the segments that classified in the same genetic clusters were concatenated and included in the analyses; the PA segment was excluded for B1, and the PB1 and NS segment was excluded for B3. Then, the seven datasets were further down sampled to eliminate sequence duplication. The A1 dataset consisted of 342 U.S. H5N1 sequences with a genetically close ancestral strain (A/whooper swan/Ireland/032444-22VIR1325-2/2021) and A2 included 23 U.S. sequences. For the reassortant genotypes the U.S. sequences used in the datasets were as follows: B1 239, B2 462, B3 119, B4 37, and B5 17.

2.5. Phylogenetic network analyses

Global network using HA and NA. A global U.S. H5N1 network was constructed using concatenated nucleotide sequences of the HA and NA coding regions. There were 30 sequences representing less than 2% of total sequences that were not assigned a genotype and were thus excluded from this analysis. Among the HA and NA concatenated gene segments, sequence duplicates with an identical sampling location for wild birds or the same premises for poultry were excluded to best represent wild bird species and collection dates by using CD-HIT Suite (Fu et al., 2012). The remaining 1031 sequences were used for the global network analysis, and each virus was annotated by color codes based on country or region, genotype, or wild bird flyway.

Genotypical network using datasets. For better phylogenetic resolution, the sequence datasets established above were analyzed by Star Contraction and median-joining network phylogenetic analysis for each major genotype separately (Bandelt et al., 1999; Forster et al., 2001). Each sequence was color coded based upon sample source (backyard flock, commercial poultry, or wild bird).

2.6. Phylogenetic analysis using BEAST

Incorporation of sampling time point, location, and sample source allowed the Bayesian time-scaled phylogenetic analysis to estimate the time of most recent common ancestor, host type transition, and spatial distribution for six major genotypes. The ancestral host state reconstruction and the asymmetric viral exchange between different host species were estimated using non-reversible continuous-time Markov chain model. A Bayesian stochastic search variable selection procedure was applied to identify the best supported individual transitions between discrete states using a Bayes factor test, as available in the SpreaD3 software v0.9.6 (Bielejec et al., 2016). The following Bayesian phylogenetic tree estimation was conducted by combining three independent Markov chains, with an effective sampling size of >200 as criteria and as calculated by Tracer version 1.5 with a 10% burn-in. Maximum clade credibility trees were generated for each dataset by using TreeAnnotator (http://www.phylo.org/index.php/tools/treeannotator.html), and were visualized using FigTree 1.4.4 (http://tree.bio.ed.ac.uk).

Time of MRCA estimation (TMRCA). To estimate the time of most recent common ancestor, Bayesian time-scaled phylogenetic trees were constructed using Bayesian Evolutionary Analysis Sampling Trees (BEAST) v1.10.4 (Suchard et al., 2018), with an uncorrelated relaxed clock model with a log-normal distribution and the Gaussian Markov random field (GMRF) Bayesian Skyride coalescent tree prior (Minin et al., 2008). Site models were set for each dataset across all MRCA estimation, based on best-fit nucleotide substitution using jModelTest2 (Darriba et al., 2012). For A1, B1, and B2 genotypes, GTR model with a proportion of invariant sites (I) and gamma-distributed rates (G) was used. For the B3 genotype, GTR + G was used. For A2, B4 and B5 GTR was used. Each Markov chain consisted of 200,000,000 steps and was sampled every 20,000 steps, and the first 10% of samples were discarded as burn-in.

Source of MRCA estimation (SMRCA). Bayesian time-scaled trees with sample source were estimated to infer virus transmission between wild bird population (wild birds) and domestic birds (backyard flock and commercial poultry), and virus transmission between different agricultural sectors (backyard flock and commercial poultry). Time-scaled trees were generated using the site model indicated above, a relaxed log-normal clock model, and constant size coalescent tree prior. An

asymmetric substitution model with the Bayesian stochastic search variable selection and a strict clock model were used for the discrete nominal category of host type (backyard flock, commercial poultry, or wild birds). Each Markov chain consisted of 100,000,000 steps and was sampled every 10,000 steps, and the first 10% of samples were discarded as burn-in.

Spatial distribution and location of MRCA estimation (LMRCA). Bayesian time-scaled trees with sampling location (U.S. state isolated) were estimated to infer spatial distribution. Time-scaled trees of were generated using the site model indicated above, a relaxed log-normal clock model, and constant size coalescent tree prior. An asymmetric substitution model with the Bayesian stochastic search variable selection and a strict clock model were used for the discrete nominal category of state as sampling location. Each Markov chain consisted of 100,000,000 steps and was sampled every 10,000 steps, and the first 30% of samples were discarded as burn-in.

2.7. In vivo pathotyping

Intravenous pathogenicity index (IVPI) tests of select H5N1 viruses was conducted in chickens according to the World Organisation for Animal Health (WOAH) Manual of Diagnostic Tests and Vaccines for Terrestrial Animals (WOAH, 2023). In brief, 0.1 mL of infectious allantoic fluid was inoculated intravenously into ten 6-week-old specific-pathogen-free chickens and the chickens were monitored for 10 days for illness and death. Isolates with IVPI >1.2 were characterized as HPAIV. The challenge studies and all experiments with live viruses were conducted in a biosafety level 3 facility at the NVSL, in accordance with approved institutional animal care and use protocols.

3. Results

3.1. Genotype classification

All the U.S. viruses analyzed have an H5 clade 2.3.4.4b HA gene and an EA lineage N1 gene (n = 1315). Genotypes were classified by geo-lineage ('A' for unreassorted EA and 'B' for EA/AM reassortants of the A1 genotype), by concatenated segment-wise genetic clusters and by phylodynamic analyses based upon ML phylogeny (Table 1 and Supplementary Table 1). Individual gene segments were classified using BLAST search and phylogenetic analyses of the reassorted genes with related sequences of AM lineage wild birds AIVs (Fig. 1).

The viruses used for the downstream analyses were classified into six major genotypes that corresponded to 92% (1205/1315) of the U.S. sequences examined. The A1 reassortant sequences that accounted for less than 2% (30/1315) of all sequence data were classified as minor genotypes. As a result of this work, a data pipeline was created to automate genotyping assignment (https://github.com/USDA-VS/GenoFLU).

3.2. Three distinct introductions of H5N1 clade 2.3.4.4b viruses into the United States were identified from December 2021 to April 2022

Global network analysis was performed using HA and NA concatenated genes from H5N1 viruses detected in wild and domestic birds from December 2021 to April 2022 (Fig. 2). The analysis shows that there were three independent H5N1 clade 2.3.4.4b virus introductions into the U.S. based on geographical and genetic distinctions: two separate H5N1 introductions into the east coast states (A1 and A2 genotypes), and another genetically distinct H5N1 introduction into Alaska (A3).

Distinct from the initial introduction (genotype A1), a second introduction of H5N1 clade 2.3.4.4b virus (A2) was detected in wild birds and a few backyard bird premises in the New England states (Fig. 2, Supplementary Fig. 3). Eight common nucleotide changes in the HA and NA concatenated genes rendered the 23 A2 viruses as a separate branch distinct from the A1 viruses and the estimated common ancestral H5N1 viruses (A/whooper swan/Ireland/032444-22VIR1325-2/2021). Likewise, each HA and NA segment of the A2 viruses, on ML phylogenetic analyses, clustered separate from the A1 viruses, with higher bootstrap value (n > 90) (Supplementary Fig. 4). Hence, the A2 viruses are a distinct genetic group that evolved separately from the A1 viruses, branching off from a common ancestor still not identified. Viruses from this genotype also caused a mortality event in harbor and gray seals in New England (Puryear et al., 2022).

A third group of H5N1 viruses (A3) (Fig. 2) was found in Alaska in April 2022. These viruses, although still H5 clade 2.3.4.4b, were genetically distinct from the two previous Eurasian genotypes. The two A3 viruses are related to the early H5N1 viruses from the Netherlands that dispersed to Europe, North America, and East Asia, including Italy, Sweden, Japan, and Canada (British Columbia); however, the closest

Table 1
Genotype for H5N1 clade 2.3.4.4b from the United States (n = 1315 all genotypes represented), including A1 viruses from Europe (n = 46) and Canada (n = 5), and A3 viruses from Japan (n = 3).

Genotype	Segment group								Constellation	# of Sequences
	PB2	PB1	PA	HA	NP	NA	M	NS		
A1	ea1	ea1	ea1	ea1	ea1	ea1	ea1	ea1	ea1|ea1|ea1|ea1|ea1|ea1|ea1|ea1	395
B1.1	am1.1	am1.1	ea1	ea1	am1.2	ea1	ea1	ea1	am1.1|am1.1|ea1|ea1|am1.2|ea1|ea1|ea1	151
B1.2	am1.1	am1.1	am1	ea1	am1.2	ea1	ea1	ea1	am1.1|am1.1|am1|ea1|am1.2|ea1|ea1|ea1	95
B2	am1.2	ea1	ea1	ea1	am1.1	ea1	ea1	ea1	am1.2|ea1|ea1|ea1|am1.1|ea1|ea1|ea1	475
B3.1	am2.1	ea1	ea1	ea1	am1.4.1	ea1	ea1	ea1	am2.1|ea1|ea1|ea1|am1.4.1|ea1|ea1|ea1	57
B3.2	am2.1	am1.2	ea1	ea1	am1.4.1	ea1	ea1	am1.1	am2.1|am1.2|ea1|ea1|am1.4.1|ea1|ea1|am1.1	79
B4	am2.2	ea1	ea1	ea1	am1.3	ea1	ea1	ea1	am2.2|ea1|ea1|ea1|am1.3|ea1|ea1|ea1	37
B5	ea1	ea1	ea1	ea1	am1.4.2	ea1	ea1	ea1	ea1|ea1|ea1|ea1|am1.4.2|ea1|ea1|ea1	18
minor	am1.1	ea1	am1	ea1	am1.2	ea1	ea1	ea1	am1.1|ea1|am1|ea1|am1.2|ea1|ea1|ea1	5
minor	am1.2	ea1	ea1	ea1	am1.1	ea1	ea1	am1.2	am1.2|ea1|ea1|ea1|am1.1|ea1|ea1|am1.2	7
minor	ea1	ea1	ea1	ea1	am1.1	ea1	ea1	ea1	ea1|ea1|ea1|ea1|am1.1|ea1|ea1|ea1	2
minor	am2.1	ea1	ea1	ea1	am1.4.1	ea1	ea1	am1.1	am2.1|ea1|ea1|ea1|am1.4.1|ea1|ea1|am1.1	1
minor	am2.2	ea1	ea1	ea1	am1.6	ea1	ea1	ea1	am2.2|ea1|ea1|ea1|am1.6|ea1|ea1|ea1	4
minor	ea1	ea1	ea1	ea1	am1.4.1	ea1	ea1	ea1	ea1|ea1|ea1|ea1|am1.4.1|ea1|ea1|ea1	4
minor	am1.1	ea1	ea1	ea1	am1.6	ea1	ea1	ea1	am1.1|ea1|ea1|ea1|am1.6|ea1|ea1|ea1	2
minor	am2.3.2	ea1	ea1	ea1	am1.5	ea1	ea1	ea1	am2.3.2|ea1|ea1|ea1|am1.5|ea1|ea1|ea1	2
minor	ea1	ea1	ea1	ea1	am1.6	ea1	ea1	ea1	ea1|ea1|ea1|ea1|am1.6|ea1|ea1|ea1	1
minor	am1.2	ea1	ea1	ea1	am1.4.1	ea1	ea1	ea1	am1.2|ea1|ea1|ea1|am1.4.1|ea1|ea1|ea1	1
minor	am2.3.1	am2	am2	ea1	am2	ea1	ea1	am1.3	am2.3.1|am2|am2|ea1|am2|ea1|ea1|am1.3	1
A2	ea2	ea2	ea2	ea2	ea2	ea2	ea2	ea2	ea2|ea2|ea2|ea2|ea2|ea2|ea2|ea2	23
A3	ea3	ea3	ea3	ea3	ea3	ea3	ea3	ea3	ea3|ea3|ea3|ea3|ea3|ea3|ea3|ea3	9

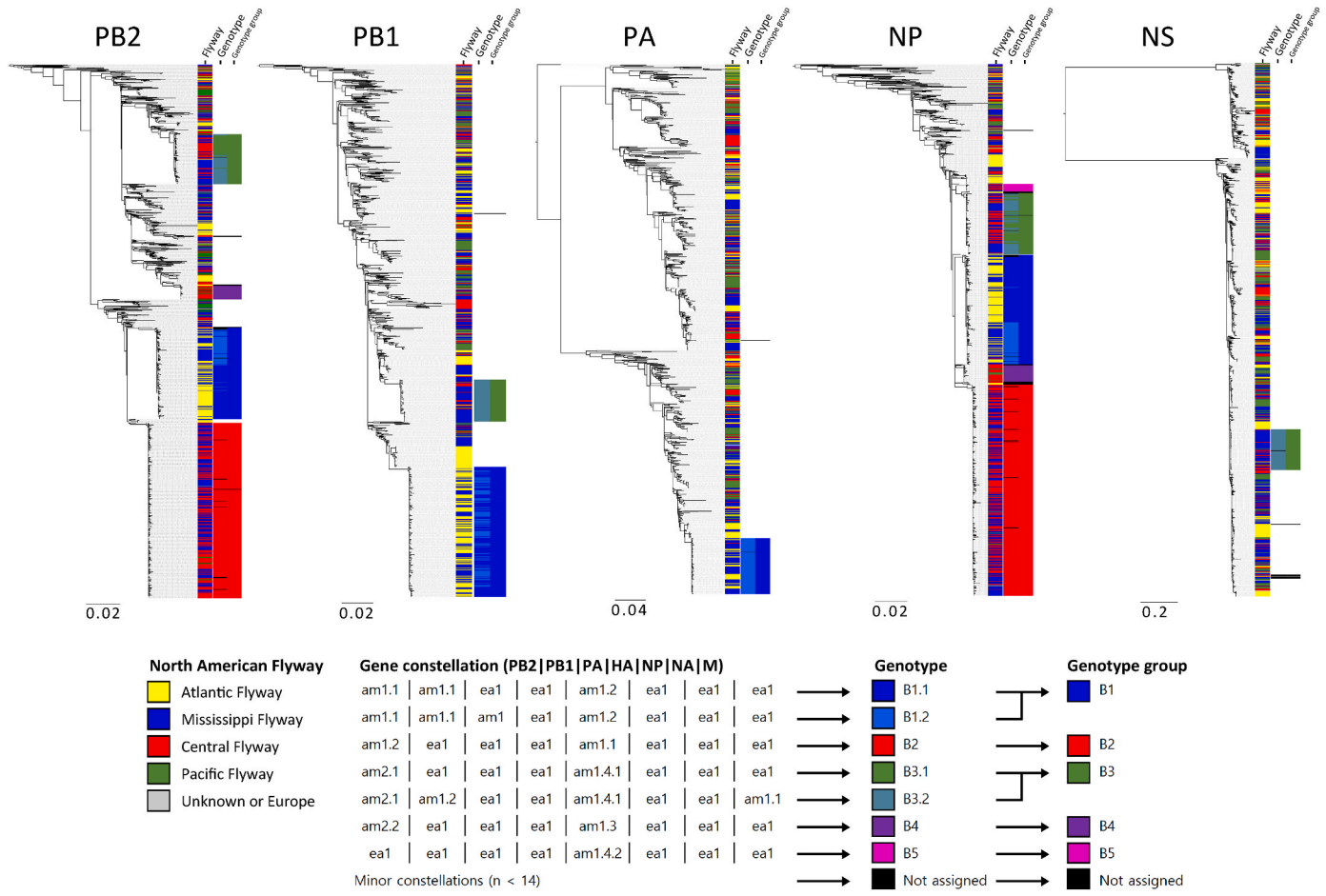


Fig. 1. Maximum-likelihood phylogenetic trees for H5N1 clade 2.3.4.4b reassortants with North American wild bird AIVs by gene segment. Reassorted segments (PB2, PB1, PA, NP, and NS) of H5N1 clade 2.3.4.4b with similar North American wild bird AIVs were used for maximum-likelihood tree construction. The trees are color-coded by sampling location (flyway) and genotype.

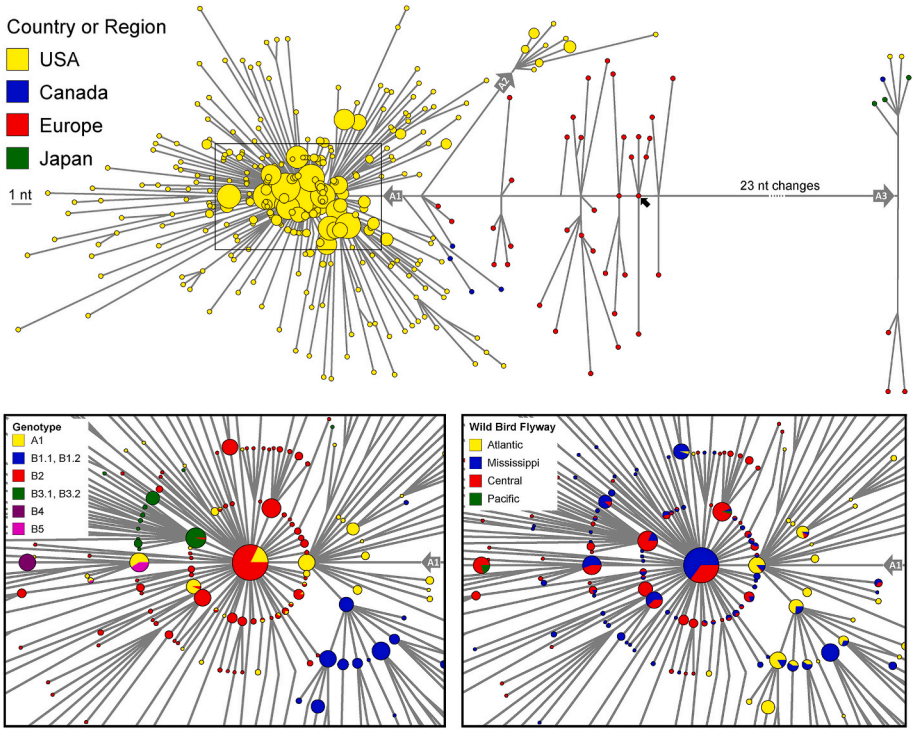


Fig. 2. Phylogenetic network of H5N1 clade 2.3.4.4b using HA and NA concatenated genes. Phylogenetic network of H5N1 viruses sequenced in this study with other publicly available sequences of high identity was constructed using concatenated coding regions of HA and NA segments. Identical sequences or star-like shaped clusters where a common ancestor branches to each descendent with branch length equal to one were contracted to single node using Star Contraction integrated in Network v10.2. The size of node correlates with the number of contracted sequences. The black arrow represents an H5N1 clade 2.3.4.4b virus from the Netherlands in October 2020. The three distinct introductions of H5N1 clade 2.3.4.4b viruses are denoted with A (A1, A2, A3). Reassortant genotypes of A1 with North American (AM) wild bird AIVs are represented with B (B1, B2, etc.). The dotted region in the branch reaching to A3 indicate 23 nt changes. The top network is colored by location to country level (USA, Canada, and Japan) or region (Europe). The lower left panel is colored by genotype A1 and its reassortants; the lower right panel is colored by flyway.

genetic matches were from wild bird viruses from Japan and Canada.

3.3. Genetic reassortment with North American wild bird LPAIVs

Evidence of reassortment involving up to five internal gene segments (PB2, PB1, PA, NP, and NS) were found between the A1 genotype viruses and AM wild bird-origin LPAIVs (Supplementary Table 3). Reassortant viruses represented 72% of the U.S. viruses examined (942/1315), and all retained the H5 2.3.4.4b HA (A1 genotype), EA N1, and EA matrix genes, with an AM NP gene from different AM NP lineages. As of April 2022, no reassortants of genotype A2 or A3 had been identified. To further examine the dynamics of virus spread across the U.S., H5N1 reassortant viruses were assigned to the major waterfowl flyway where each was detected, namely the Atlantic (ATL), Mississippi (MISS), Central (CEN), and Pacific (PAC). These flyways are delineated along the longitudinal axis of North America and correspond to the seasonal migratory routes of wild aquatic birds (USDA-APHIS, 2023d). Movement of birds along the various flyways has been shown to be an important route of large-scale geographical spread of AIVs (Fourment et al., 2017) and thus can provide insight into the origin and subsequent spread of genotypes.

AM PB2 genes: Classified in six clusters: am1.1, am1.2, am2.1, am2.2, am2.3.1, am2.3.2 (Supplementary Table 3). The am1.1 and am1.2 of PB2 were closely related, with high between-group nucleotide identity (98.4%). However, two independent monophyletic groups were observed with distinct spatial distribution: am1.1 spread primarily along the ATL (n = 146) and MISS (n = 104) (CEN = 3 and PAC = 0), whereas am.1.2 spread mostly along CEN (n = 241) and MISS (n = 234) (ATL = 1, and PAC = 7). Another genetically distinct North American PB2, denoted as am2, was mostly found in the MISS (n = 75) and CEN (n = 90) flyways (ATL = 6, and PAC = 10). This group was further subdivided into four subgroups am2.1, am2.2, am2.3.1, and am2.3.2 based on phylogenetic and spatial distribution, which indicated that the reassortment events took place in different locations.

AM PB1 genes: Classified in three clusters am1.1, am1.2, am2 (Supplementary Table 3). The am1.1 and am1.2 were found in distinct geographical locations indicating two separate reassortment events and virus spread according to wild bird movements. The highest between-group sequence identity was 96.2% and was found between am1.1 and am1.2. The am1.1 and am1.2 subclusters had two patterns of spatial distribution: am1.1 tended to spread along the ATL and MISS flyways (ATL = 144, MISS = 100, and CEN = 1), whereas am1.2 reached the PAC flyway as the virus spread westward (ATL = 1, MISS = 61, CEN = 15, and PAC = 2). One virus from New York (A/Pheasant/New York/22-008760-010/2022) had a PB1 gene (designated am2) that is phylogenetically distinct from the am1.1 and am1.2 clusters.

AM PA genes: The majority of viruses retained the ea1 PA gene, but two AM clusters were identified: am1 and am2 (Supplementary Table 3). The am1 and am2 clusters were both phylogenetically distinct and shared low genetic identity, with the highest similarity of only 88.5%. Most AM lineage PA genes belong to the am1 cluster with distribution centered around the Midwest. One virus from New York had unique PA and PB1 genes, both designated am2.

AM NP genes: All reassortants (942/942) had an AM NP gene and were classified into eight clusters: am1.1, am1.2, am1.3, am1.4.1, am1.4.2, am1.5, am1.6, and am2 (Supplementary Table 3). The NP gene segment had the greatest number of reassortments with 8 distinct clusters, 7 of which were subclusters of the am1 group. The highest between-group genetic identity in am1 ranged from 98.2 to 99.0%, with distinct monophyletic clusters. The two most frequent AM lineage NP clusters, am1.1 (n = 483) and am1.2 (n = 251), showed well-defined spatial distribution. The am1.2 cluster spread primarily along ATL (n = 146) and MISS (n = 104) flyways (CEN = 1, and PAC = 0), whereas am1.1 spread along the CEN (n = 234) flyway (ATL = 1, and PAC = 7). Along the PAC flyway, am1.3 was most dominant particularly in the northwestern states and Alaska. Similar spatial distribution and cluster formation with high bootstrap values (n = 89) on the phylogenetic tree indicates that subclusters am1.4.1 and am1.4.2 originated from a common ancestor but diverged into different populations in wild birds. Other minor subgroups am1.5 and am1.6 were also found as of April 2022. The am1.6 was detected infrequently in two distant flyways (ATL = 4, and CEN = 3) and across three different first order genotypes (Fig. 3). Similar to the PA gene, the am2 cluster was represented by a single virus (the same virus from New York with the unique am2 group of PB1 and PA).

AM NS genes: The majority of the viruses maintained the ea1 NS gene. Three subclusters of AM lineage NS genes were identified as of April 2022: am1.1, am1.2, and am1.3 (Supplementary Table 3). These three subclusters share high between-group genetic identity (97.9–98.7%). The am1.1 was detected in the CEN and MISS flyways, generating one of the six major genotypes, B3. The genotype of seven viruses with an NS gene from am1.2 is B2 except at the NS segment, indicates that reassortants with am1.2 genotype are the result of secondary reassortment of B2 genotype viruses. One virus with am1.3 was the same virus from New York with the unique am2 groups of PB1, PA, and NP.

No reassortments of the HA, MP, or NA gene segments were observed during the time period described. Rare AM NA reassortants have been detected since April 2022 (USDA-APHIS, 2023a).

3.4. Closest wild bird LPAIV gene donors

Potential wild bird LPAIVs that served as the closest gene donors for the reassortants were traced according to the genotype and phylogenetic relationships (Fig. 3). The B1 and B2 genotypes had a shared potential donor namely the H7N6 viruses from ruddy turnstones (*Arenaria interpres*) from Delaware Bay in 2020 (A/ruddy turnstone/Delaware Bay/210–212/2020-like). Because the B1 and B2 genotypes differ in spatial pattern and phylogenetic clustering, the descendants from the index donor viruses were subdivided into two descendants-I and –II, respectively. The closest donors for B3 genotype were H4N6 and H6N5 viruses from mallards (*Anas platyrhynchos*) in Alberta in 2019 (A/mallard/Alberta/469–588/2019-like). Likewise, the potential donors for B4 were H1N1 and H3N8 viruses from mallards in Alberta in 2019 (A/mallard/Alberta/147–428/2019-like). Because NP was the only reassorted segment in B5 and was closely related to that of B3, a potential donor for the NP reassortment in B5 could be a descendant of A/mallard/Alberta/469–588/2019-like viruses. The other minor reassortants were classified either into subsets of the major reassortants or as independent episodic reassortants. One virus had segments from both B2 (PB2) and B3 (NP) genotypes. The other three types of episodic reassortants indicate independent and/or secondary reassortment events; these are reassortants harboring the NP gene of am1.6 group as donor, two viruses from snow geese (*Anser caerulescens*) in North Dakota with distinct PB2 and NP genes, and one virus from a pheasant (*Phasianus colchicus*) in New York with a rare gene constellation that did not have clear relationships with other genotypes. Notably, the unreassorted segments (HA, NA, and M) of the pheasant virus isolated from a backyard flock in New York showed high sequence identity with that of A1 viruses from backyard birds in Maine, indicating a genotype A1 virus introduction to backyard birds in Maine and potential transmission to New York via some backyard connection, with coincidental wild bird virus reassortment.

3.5. Transmission dynamics by time, sample source, and location estimation of MRCA by first order genotype

Spatial transmission in relation with time, source, and location of most recent common ancestor (MRCA). To estimate when and where each genotype of H5N1 viruses originated from in the U.S., the six major genotypes were analyzed using Bayesian methods. The genotype A1 and reassorted genotype B subsets make up the six major genotypes and were prioritized for analysis (A1, B1, B2, B3, B4, and B5). The virus movements (called transitions) between states across the U.S. were mapped

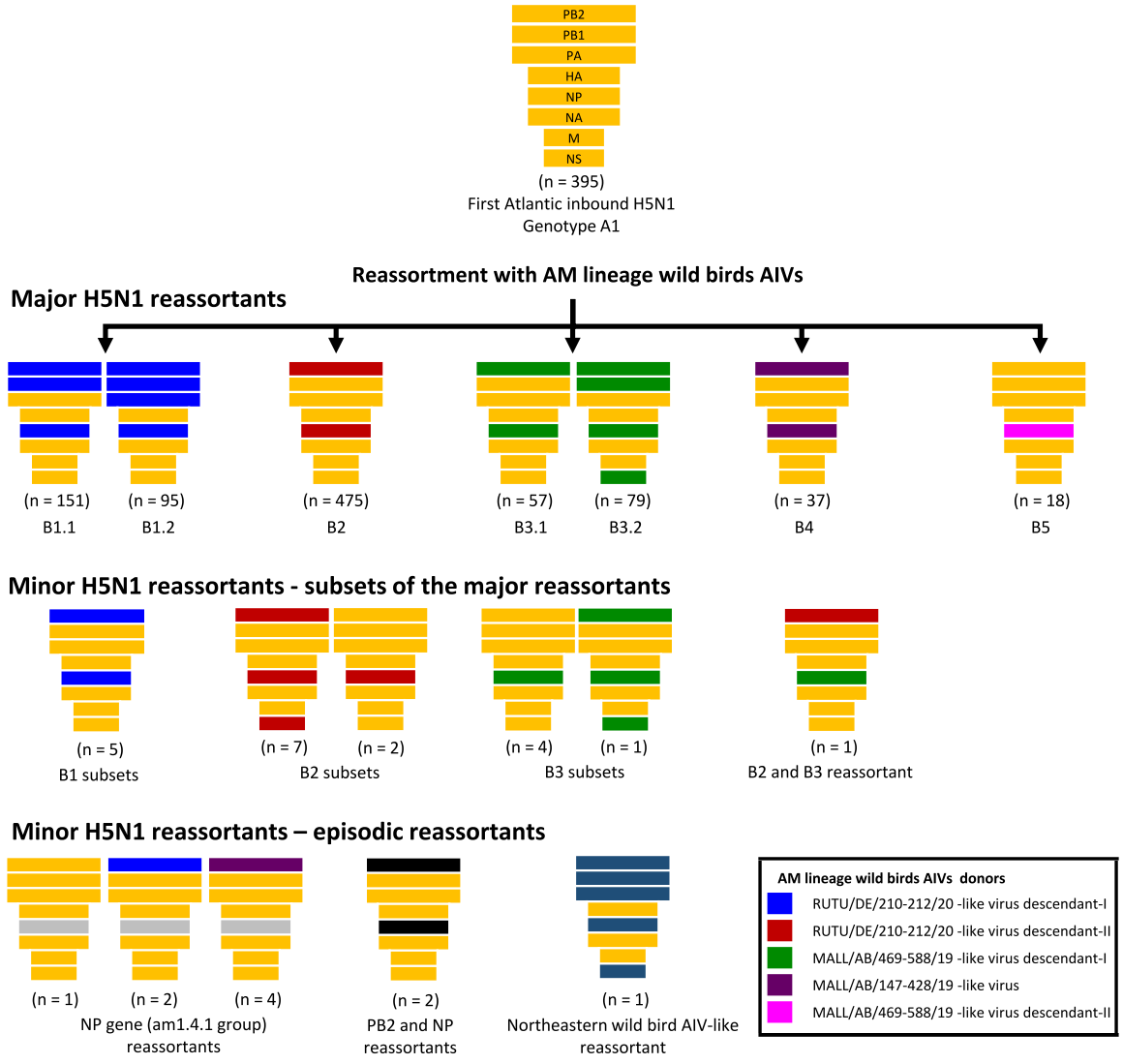


Fig. 3. Emergence of H5N1 clade 2.3.4.4b reassortants Schematic of eight viral gene segments are visualized and colored according to the phylogenetic tree analyses and classification. All reassortants arose from the initial introduction of the H5N1 A1 genotype (orange). Seven genotypes of H5N1 reassortant HPAI viruses were generated by internal gene reassortment with five North American (AM) wild bird LPAIVs including by two descendants of A/ruddy turnstone/Delaware Bay/210–212/2020 (H7N6, RUTU/DE/210–212/20-like descendent-I and –II, blue and red respectively), and A/mallard/Alberta/469–588/2019 (H4N6 and H6N5, MALL/AB/469–588/19-like descendent-I and –II, green and fuchsia respectively), with A/mallard/Alberta/147–428/2019 (H1N1 and H3N8, MALL/AB/147–428/19,purple). The remainder were regarded as minor reassortants wherein six types represented further reassortants of the major reassortants (B1–B5), and five were episodic reassortants.

according to major genotype with the estimated number of virus transitions represented by median Markov Jump counts (Fig. 4). Estimated values related to MRCA are in Table 2. Fig. 5 illustrates the time series incidence of the major H5N1 genotypes sequenced in this study.

For genotype A1, phylodynamic analysis estimated that the A1 viruses first arrived in northeast ATL flyway states in November 2021 [median time MRCA (TMRCA), November 6, 2021] even though the first H5N1 detection in the U.S. was in wild birds in South Carolina. The A1 viruses then rapidly spread along the ATL flyway to North Carolina and South Carolina, and south to Florida along the Atlantic coast. At the same time, the A1 viruses spread via the MISS and CEN flyways (Fig. 4).

The genotype B1 virus was first detected in Georgia in January 2022 (median TMRCA of January 1, 2022), with gene reassortments either in PB2 and NP or in PB2, PB1, and NP (Fig. 3). After which, the B1 viruses caused a sustained wild bird outbreak in Florida (Fig. 4). B1 viruses also moved into northern midwestern states, causing a second major wild bird outbreak in Michigan.

Once the A1 genotype viruses spread into the MISS and CEN flyway states, reassortment was detected in PB2 and NP for B2, and either in PB2 and NP or in PB2, PB1, NP, and NS for B3 (Fig. 3). North Dakota was estimated to be location MRCA (LMRCA) for both B2 and B3 in January 2022 (median TMRCA of January 15, 2022, and January 17, 2022, for B2 and B3 respectively). These two reassortant viruses spread mostly to neighboring states and less commonly to distant Atlantic states (New York and Maryland for B2 and B3, respectively).

Reassortment of A1 and wild bird AIVs in PB2 and NP gene gave rise to the genotype B4 during March 2022 (median TMRCA of March 5, 2022). The MRCA of B4 was estimated to be a wild bird in Montana. The B4 viruses then further reached other states including Wyoming, Idaho, and Alaska, through the PAC flyway. Of note, the B3 and B4 was also found in wild birds in Alaska, where B4 was detected in a backyard farm. This demonstrates the long-distance virus spread of the H5N1 across North America.

The TMRCA of the B5 genotype was estimated to be November 2021, which is earlier than the first detection of this genotype on April 3, 2022. This finding is likely due to the small number of samples having the B5

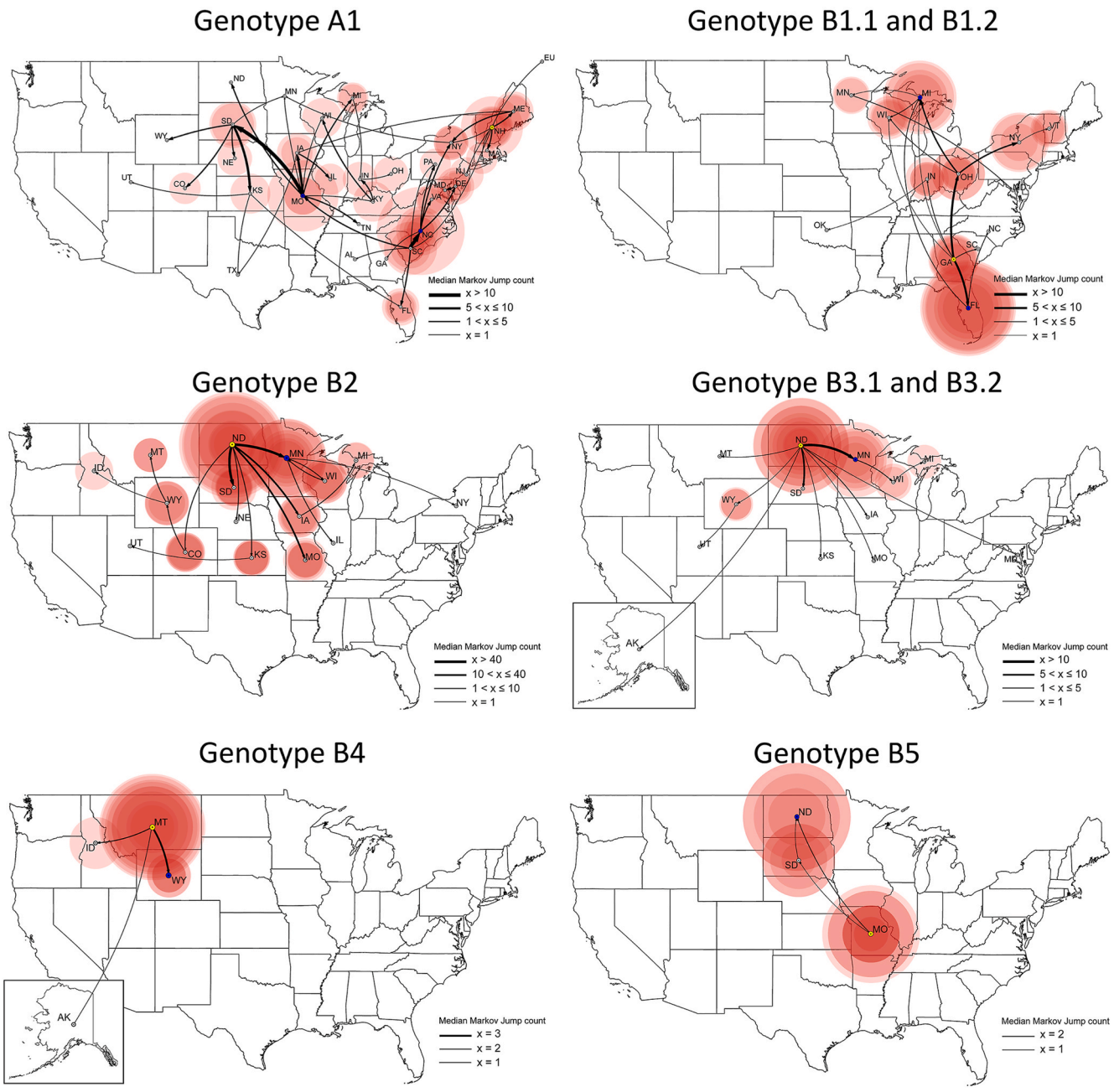


Fig. 4. Spatial diffusion of US H5N1 clade 2.3.4.4b viruses by genotype. Each panel is phylogeographic representation of the A1 genotype and its reassortants. Arrows indicate the direction of virus transition as estimated by a Bayesian stochastic search variable selection. The thickness of arrows represents the median Markov jump count/s for all the transitions between geographical location states along phylogenetic branches. The size and intensity of the red circles is proportional to the frequency of detections and number of genotypes found. The temporal formation and extinction of the red circles were superimposed during the surveillance time. The LMRCA (yellow dots) or the location (blue dots) represent the first or second highest mean Markov rewards. The spatial diffusion of genotype A2 is shown in Supplementary Fig. 3.

AK, Alaska; AL, Alabama; CT, Connecticut; CO, Colorado; DE, Delaware; EU, Europe; FL, Florida; GA, Georgia; IA, Iowa; ID, Idaho; IL, Illinois; IN, Indiana, KS, Kansas; KY, Kentucky; MA, Massachusetts; MD, Maryland; ME, Maine; MI, Michigan; MN, Minnesota; MO, Missouri; MT, Montana; NC, North Carolina; ND, North Dakota, NE, Nebraska; NH, New Hampshire; NJ, New Jersey; NY, New York; OK, Oklahoma; PA, Pennsylvania; SC, South Carolina; SD, South Dakota; TN, Tennessee; UT, Utah; VA, Virginia; VT, Vermont; WI, Wisconsin; WY, Wyoming.

genotype and its lack of monophyletic ancestry.

The second introduction of H5N1 to the U.S. by a distinct HA group virus was identified in the ATL flyway into the Northeast states around January 2022 (median TMRCA of January 18, 2022). These A2 genotype viruses remained localized within Northeast states including Connecticut, Maine, and Massachusetts, and New York as of April 2022. The estimated LMRCA was Connecticut or Massachusetts, with detections from wild birds and sporadic backyard flocks.

Transmission among host types and potential lateral and common source transmission. Estimated rate and number of host transitions among three different host types (wild bird [WB], commercial poultry [CP], and backyard flock [BF]) were modelled using the Bayesian stochastic search. Here, a virus host type was considered as a discrete nominal category, and the switches in virus host type by genotype were inferred through Bayesian phylogeny. Overall, the major host type with the highest estimated mean Markov reward time was WB across all viruses,

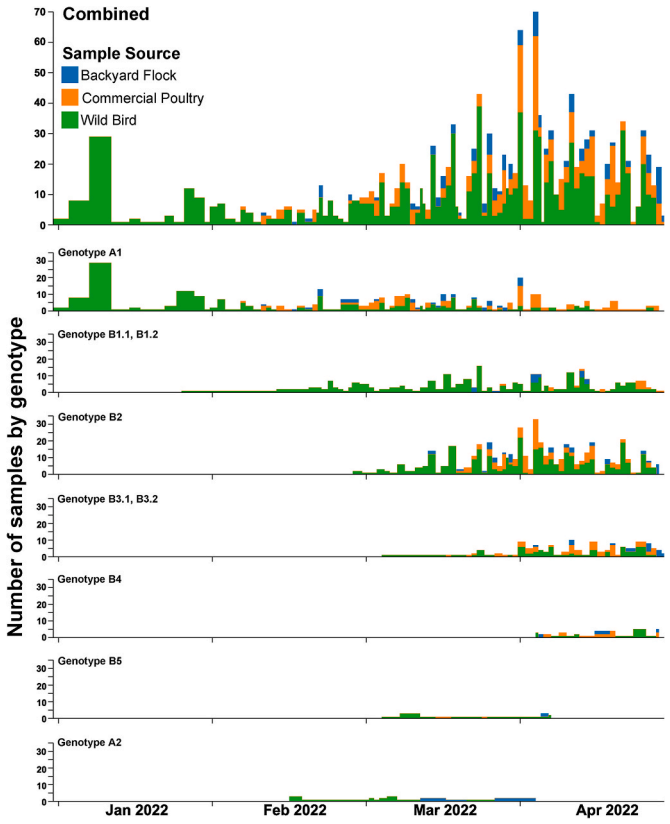


Fig. 5. Time series of US H5N1 clade 2.3.4.4b detections by genotype. The frequency of the H5N1
detections are represented over time, color coded by source (backyard flock, commercial poultry, and wild birds), and displayed by genotype (excluding A3 for which only two viruses were identified: A/bald eagle/Alaska/22-013001-001/2022 and A/bald eagle/Alaska/22-013831-001/2022 collected on 26 and 27 April 2022 respectively).

indicating that the H5N1 viruses had largely been sustained in wild bird populations regardless of virus genotype (Table 2). Next, CP was inferred to have the second highest Markov reward time specifically for A1, B2, and B3 genotypes, whereas relatively low and insignificant values of CP and BF were estimated for B1, B4 and B5. The differences in the reward time indicates that WB-to-CP transition was more frequent for the A1, B2, and B3 compared to B1, B4, and B5. Indeed, the WB-to-CP transition rates and corresponding Markov jump count for A1 (n = 27), B2 (n = 62), and B3 (n = 21) were higher than B1 (n = 7), B4 (n = 4), and B5 (n = 2) (Table 3).

Phylogenetic network analyses of major genotype datasets indicated that most of WB-to-CP transition represent sporadic, point-source outbreaks. However, some host type transitions for CP and BF indicate lateral or common source transmission (Supplementary Fig. 5). The longest such transmission chain was estimated to last at least one month, involving four different premises of turkey (*Meleagris gallopavo domesticus*) farms in Indiana during February and March of 2022. Sustained transmissions involving the greatest numbers of premises were four chicken premises in Maryland and three chicken premises in Delaware. Both events were caused by A1 viruses. Although evidence of farm-to-farm spread was identified in some cases, the majority of cases were consistent with point-source introductions or common wild bird source exposures. Of note, no host type transitions from CP or BF to WB were identified across any genotypes.

Potential lateral or common source transmissions between domestic birds (CP and BF) were inferred by examining the clustering of viruses from CP and BF on the phylogenetic network analyses for each genotype. Estimated host type transitions between CP and BF were quite rare and only observed in three genotypes [A1 (n = 1), B2 (n = 1), and B4 (n = 2)]. Potential farm-to-farm or common source transmission was estimated to occur in 13 states involving A1, B1, B2, and B3 viruses. Of note, transmissions to CP and BF during the same time period involving different genotypes occurred in Indiana, Iowa, Michigan, and Minnesota.

3.6. Pathotyping of H5N1 viruses and variations in HA cleavage site

A total of nine different HA cleavage site motifs were identified in the U.S. HA sequences examined (n = 1315). The most frequently observed HA cleavage site sequenced in this study (PLREKRRKR/GLF, 1297 out of 1315) was consistent with the previously known multi-basic cleavage site of H5 clade 2.3.4.4 H5Nx HPAIV (Supplementary Table 4). Of the other eight minor motifs, two were identical to previously reported motifs (PLRERRRKR/GLF and PQREKRRKR/GLF) and six were new motifs with no insertions or deletions. The eight motifs contained five basic amino acids at the −1 to −5 positions while one minor motif had four basic amino acids due to lysine-to-glutamic acid mutation

Table 2
The phylogenetic groups, major reassortants, and estimated MRCA of HPAI clade 2.3.4.4b in United States from December 2021 to April 2022.

First order genotype	Median TMRCA (Date of the first detection)	SMRCA (Posterior)	Estimated mean Markov Rewards of Sample source (sequence count)			LMRCA (Posterior)	Estimated mean Markov Rewards of Location (sequence count)		Total sequences count
			WB	CP	BF		First	Second	
A1	6 Nov 21 [31 Oct 21–8 Nov 21] (12/30/2021)	Wild bird (1.0)	17.25 (198)	3.58 (115)	1.86 (35)	NH (0.5787)	NC 3.95 (68)	MO 3.12 (22)	348
B1	1 Jan 22 [18 Dec 21–12 Jan 22] (1/25/2022)	Wild bird (1.0)	18.51 (212)	0.49 (15)	0.51 (18)	GA (0.9934)	FL 7.28 (100)	MI 3.12 (55)	245
B2	15 Jan 22 [1 Jan 22–26 Jan 22] (3/2/2022)	Wild bird (1.0)	27.76 (308)	4.64 (128)	1.35 (38)	ND (0.9601)	ND 17.7 (120)	MN 8.5 (142)	474
B3	17 Jan 22 [23 Dec 21–6 Feb 22] (3/4/2022)	Wild bird (1.0)	9.15 (66)	1.78 (53)	0.73 (17)	ND (0.8816)	ND 7.54 (43)	MN 3.76 (62)	136
B4	5 Mar 22 [6 Feb 22–22 Mar 22] (4/4/2022)	Wild bird (1.0)	1.16 (17)	0.16 (14)	0.18 (6)	MT (0.9341)	MT 1.38 (24)	WY 0.17 (5)	37
B5	28 Nov 21 [15 Nov 19–26 Feb 22] (3/4/2022)	Wild bird (1.0)	5.36 (13)	0.73 (2)	0.84 (2)	MO (0.5)	ND 1.55 (8)	MO 1.12 (6)	17
Na	18 Jan 22 [21 Dec 21–5 Feb 22] (2/16/2022)	Wild bird (1.0)	1.34 (16)	Na (0)	0.3 (7)	CT or MA (0.3502, 0.3272)	MA 2.23 (10)	ME 0.48 (8)	23
Na	Nd (4/26/2022)	Nd	2	0	0	Nd	Nd (2, AK)	Nd	2

Na, not available. Nd, not done. WB, wild bird; BF, backyard flock; CP, commercial poultry.
AK, Alaska; CT, Connecticut; FL, Florida; GA, Georgia; MA, Massachusetts; ME, Maine; MI, Michigan; MN, Minnesota; MO, Missouri; MT, Montana; NC, North Carolina; ND, North Dakota, NH, New Hampshire; WY, Wyoming.

Table 3
U.S. H5N1 clade 2.3.4.4b estimated transmission between host types as represented by the mean actual rate of host type transition and median Markov Jump (MK; estimated number in parenthesis) and listed by major genotype.

Genotype	Estimated transition rate between host types (MK jump count) [a]						States where potential lateral or common source spread occurred (number of premises) [b]	Total number of clusters [b]
	WB to CP	WB to BF	CP to BF	BF to CP	CP to WB	BF to WB		
A1	1.905 (27)	1.24 (17)	0.36 (1)	0.056 (0)	0.04 (0)	0.288 (0)	IN (4), ME (4), DE-MD (7), IA (2), NC (9), PA (7)	6
B1	1.556 (7)	1.392 (6)	0.271 (0)	0.212 (0)	0.314 (0)	0.193 (0)	MI (2), IN (3)	2
B2	2.364 (62)	0.724 (18)	0.141 (1)	0.022 (0)	0.041 (0)	0.02 (0)	SD (2), IA (4) [c], MN (14) [d], WY (3), MI (2), ID-MT (3)	11
B3	2.306 (21)	0.806 (7)	0.237 (0)	0.956 (0)	0.135 (0)	0.084 (0)	MN (12) [e]	2
B4	1.173 (4)	0.604 (1)	0.877 (2)	0.835 (1)	0.279 (0)	0.197 (0)	Nd	Nd
B5	1.109 (2)	0.786 (1)	0.614 (0)	0.402 (0)	0.629 (0)	0.407 (0)	Nd	Nd
Na	Na	1.033 (2)	Na	Na	Na	0.406 (0)	Nd	Nd
Na	Na	Na	Na	Na	Na	Na	Nd	Nd

Na, value not available. Nd, not detected.
DE, Delaware; IA, Iowa; ID, Idaho; IN, Indiana; MD, Maryland; ME, Maine; MI, Michigan; MN, Minnesota; MT, Montana; NC, North Carolina; PA, Pennsylvania; SD, South Dakota; WY, Wyoming.
[a] Mean actual rate of host type transition and median Markov Jump count. WB, wild bird; BF, backyard flock; CP, commercial poultry.
[b] Potential lateral or common source transmission between domestic poultry (commercial poultry and backyard flock) was identified based on clustering of sequences in phylogenetic network analyses. DE-MD and ID-MT indicate potential interstate transmission by single ancestry virus.
[c] There were two phylogenetically distinct events of potential lateral or common source spread of a B2 virus in IA accounting for a total of 4 premises.
[d] There were five phylogenetically distinct events of potential lateral or common source spread of a B2 virus in MN accounting for a total of 14 premises.
[e] There were two phylogenetically distinct events of potential lateral or common source spread of a B2 virus in MN accounting for a total of 12 premises.

(PLREKRRER/GLF). Intravenous pathogenicity index testing for 8 viruses representing 4 major genotypes (A1, B1, B2 and B3) or cleavage site changes showed little variation (2.88–3.0) among the viruses tested.

4. Discussion

The dissemination of the 2021/2022 H5N1 clade 2.3.4.4b virus in the U.S. differed markedly from the previous intercontinental spread of the 2014/2015 H5N8 clade 2.3.4.4c virus and its North American reassortant H5N2 virus, in the following aspects: (1) presence of virus in the flyways and local environment as represented by the frequency of detections in wild birds (for USDA Wildlife Services wild bird surveillance, and excluding mortality events, approximately 90 detections in 2014–2015 compared to 2800 Wildlife Services wild bird surveillance detections during 2021/2022); (2) the distribution (wild bird detections in 2015 were focused largely in the Pacific (PAC) flyway with fewer detections in the Central (CEN) and Mississippi (MISS) flyways, compared to 2022 where detections in wild birds were found across all four flyways); and (3) the number of reassortment constellations. The 2014/2015 virus introduction into the U.S. was the first documented Gs/GD lineage H5 HPAIV introduction from East Asia through Alaska during the wild bird breeding season. The wholly Eurasian H5N8 virus spread into the PAC flyway states and quickly reassorted with LPAIV NA genes of North American wild bird origin (Hill et al., 2017). Of the three reassortant viruses that emerged in the PAC flyway, only the H5N2 virus subsequently spread to Midwest states causing large scale outbreaks in turkey and chicken farms. The other reassortants and ancestral viruses were detected only in the Pacific states (Lee et al., 2018). The H5N2 virus spread to the CEN and MISS flyways by migratory birds and introduction to domestic poultry resulted in predominant farm-to-farm transmission (Hicks et al., 2020). During the 2014/2015 event, no H5 clade 2.3.4.4c viruses were detected in the ATL flyway, and approximately 24 wild bird species were affected.

In contrast, the 2021/2022 H5N1 2.3.4.4b viruses dispersed across all four flyways and affected 159 wild avian species from January 12 to December 31, 2022 (USDA-APHIS, 2023c), including the 68 species confirmed in this study, as well as sporadic spill over into wild mammals (USDA-APHIS, 2023b); these numbers continue to grow. The majority of poultry events (19 cases) at least 85%, represent sporadic/point-source introductions across the U.S. with rare secondary farm-to-farm spread largely due to greatly reduced time to depopulation by U.S. veterinary authorities compared to the 2014/2015 outbreak (USDA-APHIS, 2022). The spread of the virus across all four migratory bird flyways appears to be a major factor in ongoing virus circulation. As the H5N1 virus spread westward and southward, extensive genetic reassortment occurred with existing North American wild bird LPAIVs, generating 19 different gene constellations composed of six major genotypes and 11 minor genotypes during the period examined (December 2021–April 2022). In comparison to the clade 2.3.4.4c virus that swapped its NA gene soon after introduction during the 2014/2015 event, the N1 associated with this current H5 clade 2.3.4.4b virus has remained predominantly the same, likely indicating greater viral fitness of this H5N1 gene combination in wild birds compared to the prior H5N8 subtype predominant in Europe during 2021 (EFSA et al., 2021). Reassortment of the NA gene has been extremely rare and often difficult to tease out of a mixed infection in wild birds. The other gene segment with no evidence of reassortment has been the matrix gene. The matrix gene has been shown to interact with the HA gene, such that the cooperative interaction is necessary for efficient replication (Scholtissek et al., 2002; Webster et al., 1989), and has been also shown to increase contact transmission of the 2009 H1N1 human influenza virus (Campbell et al., 2014).

In this study, genetic clusters of each segment of the H5N1 viruses were classified by geolineage and ML phylogeny, and then the genotypes were assigned based on gene constellation composed of sets of segment-based genetic clusters. This genotype classification allowed for inclusion of as many viruses as possible while simplifying the downstream analyses; however, rare subsets within the major reassortants were excluded. These minor/episodic viruses were found in the B1, B2, and B3 genotypes, and some had mixed gene constellations from B2 and B3. These observations indicate that the PB2, PB1, PA, NP, and NS genes of the H5N1 clade 2.3.4.4b virus evolved by forming transient genome constellations as seen in wild bird AIV evolution, whereas the HA, NA, and M gene appear more stable, similar to what is seen among mammalian-adapted influenza A viruses (Dugan et al., 2008).

Several intercontinental waves of Gs/GD H5 lineage virus spread have been caused by the H5 clade 2.3.4.4 viruses since 2014 (Sims et al., 2017), and this clade continues to undergo reassortment in Europe, Asia, and the U.S., in particular the polymerase complex genes. For example, reassortment with gull-adapted H13 LPAIV circulating in seabirds was noted in Europe (EFSA et al., 2022a). Similar trends were also observed with the 2021/2022 U.S. H5N1 viruses, in terms of the reassorted gene segments and the contribution of seabird LPAIVs in these reassortant viruses. Based on the host type transition analyses in this study, the B2 viruses have a greater number of estimated WB-to-CP transitions and are involved in more potential lateral/common source spread clusters between domestic birds. The frequency of spillover to poultry appears to differ among genotypes and follow-up studies would be beneficial to determine whether virus fitness in wild birds differs among reassortant viruses and to assess changes in infectivity for domestic birds.

The 2021/2022 H5N1 viruses caused a larger outbreak in wild birds than the previous 2014/2015 H5N2 and H5N8 viruses, not only based on the number of birds affected but also in the number of species. In the 2014/2015 outbreak, wild bird species affected primarily belonged to wild waterfowl and, slightly less frequent, raptor species (hawk, owl, and falcon) (Bevins et al., 2016). The 2021/2022 H5N1 outbreak is still centered on waterfowl species such as mallard but has also affected a substantially greater variety of species including seabirds, raptors, and scavenging species such as vultures (USDA-APHIS, 2023c) (Table 4 and Supplementary Table 2). It has been clearly determined that the long-distance virus spread of the previous 2014/2015 H5N8 clade 2.3.4.4c virus was associated with wild waterfowl migration along flyways (Bevins et al., 2016; Lee et al., 2015). Although some raptor species are migratory, it is unclear if they contribute to long-distance spread of HPAIV since they appear highly susceptible to HPAIV infection (Bertran et al., 2012; Hall et al., 2009). Additional experimental studies would be of value to determine the infectious dose in raptors and whether subclinical infection occurs. Seabirds are also susceptible to AIV infections (Lang et al., 2016), and there is evidence that indicates they were also involved in long-distance transmission of AIVs along the ATL flyway (Poulson et al., 2020). Indeed, A1, A2, and B1 H5N1 viruses that spread along the ATL flyway were isolated from pelicans and seabirds including gulls, terns, and sanderlings.

Using the full genome sequence data obtained through wild bird and poultry surveillance activities, we were able to delineate spatial diffusion of the 2021/2022 H5N1 viruses by genotype. In general, spatial diffusion analyses reveal that A1 viruses that were first introduced to the U.S. through the Northeast states took two separate paths: (1) a longitudinal and southbound path from Northeast states to Florida, and (2) latitudinal and westward path from Northeast states to Midwest states. The first southbound spread was involved in the emergence of B1, and the second westward spread contributed to the generation of B2, B3, B4, and B5, with B3 and B4 reaching to the Pacific states and even Alaska. According to the date of the first detection of each genotype and estimated TMRCA, reassortant genotypes emerged in order of B1, B2, B3, and B4. For B2 and B3, the gap between the first date of detection and TMRCA may indicate that the reassortants emerged somewhere in Canada, and subsequently spread southward by wild bird migration along the CEN flyway, as determined by LMRCA estimation. Considering all the spatial diffusion analyses together, the major genotypes A1, and B1 to B5, swept through North America in approximately 4 months. Such longitudinal intracontinental spread of AIVs across North America via wild migratory birds has also been previously demonstrated in a study of N6 NA sequences (Mine et al., 2019). A caveat of the current study is that phylogenies rely on the availability of viral genome sequences; there are several wild bird efforts, each with different biases to leverage resources most efficiently, and none represent the entire wild bird population.

In this study, we found three distinct introductions of the H5N1 clade 2.3.4.4b viruses (A1, A2 and A3) into the U.S., and more are expected to be discovered if the 2.3.4.4b HPAIV continues to circulate in European and Asian wild aquatic birds. Introduction of the A2 viruses originated from a common ancestor H5N1 in Europe and the data indicate that the diversification of the A1 and A2 genotypes occurred prior to introduction to the U.S., likely while the viruses moved across the Atlantic via Iceland, Greenland/Arctic, or pelagic route (Caliendo et al., 2022). The A3 viruses have high sequence identity with H5N1 clade 2.3.4.4b viruses from Japan as early as January 2022 (Isoda et al., 2022) suggesting that the two Alaskan viruses detected in April 2022 could represent another distinct introduction into the PAC flyway via Beringia (Isoda et al., 2022). Since the A3 viruses had a significant number of nucleotide changes, a distinctive trajectory of evolution and/or transmission is anticipated as compared to their counterparts (A1 and A2 viruses) by a previously unrecognized circumpolar route, distinct from the recognized patterns of the H5 HPAI spread via Beringia or Iceland (Lee et al., 2015; Gunther et al., 2022).

Previous studies propose that key factors driving virus diversity and seasonality of LPAIVs in wild birds include environmental stability of the virus and the annual influx of large numbers of naïve juvenile birds (Roche et al., 2014; van Dijk et al., 2014). Indeed, experimental studies show that LPAIVs remained viable for more than 7 months in U.S. northern wetlands (Ramey et al., 2020). Although the 2014/2015 H5 clade 2.3.4.4c virus was not sustained in North American flyways (Lee et al., 2017), the current 2021/2023 H5N1 clade 2.3.4.4b viruses have become widespread across all four U.S. flyways. Potential endemicity with seasonality cannot be ruled out as evidenced by the ongoing and year-round persistence in the Eurasian flyways since 2020, with seasonal peaks in late fall, winter, and early spring EFSA et al. (2022b). Additionally, the 2021/2023 H5N1 clade 2.3.4.4b viruses have caused mortality events in both terrestrial and aquatic sea mammals in many countries indicating that the potential for enzootic transmission, which

Table 4
Wild birds affected by H5N1 clade 2.3.4.4b by bird species order by genotype A1 to B3.

Order	Detection frequency by host species		Number of viruses by first order genotype				Total
	First	Second	A1	B1	B2	B3	
ANSERIFORMES	Snow Goose, *Anser caerulescens* (144)	Canada Goose, *Branta canadensis* (46)	157	68	182	37	444
ACCIPITRIFORMES	Bald Eagle, *Haliaeetus leucocephalus* (86)	Black Vulture, *Coragyps atratus* (64)	17	121	71	10	219
STRIGIFORMES	Great Horned Owl, *Bubo virginianus* (52)	Snowy Owl, *Bubo scandiacus* (7)	9	9	38	7	63
PASSERIFORMES	American Crow, *Corvus brachyrhynchos* (15)	Magpie, Pica *hudsonia* (2)	1		6	11	18
GALLIFORMES	Wild Turkey, *Meleagris gallopavo* (9)	Pheasant, *Phasianus colchicus* (4)	6		8		14
CHARADRIIFORMES	Herring Gull, *Larus argentatus* (7)	Na	2	8	1		11
PELECANIFORMES	American White Pelican, *Pelecanus erythrorhynchos* (5)	Na	7	2	2		11
APODIFORMES	Black Swift, *Cypseloides niger* (2)	Na	2				2
FALCONIFORMES	Peregrine Falcon, *Falco peregrinus* (2)	Na			1	1	2
SULIFORMES	Cormorant, unspecified(2)	Na		2			2
GAVIIFORMES	Common Loon, *Gavia immer* (1)	Na		1			1
GRUIFORMES	Sandhill Crane, *Grus canadensis* (1)	Na			1		1

Na, not available. No second most frequently detected host species or unable to determine due to incomplete species data.

warrants close monitoring from a One Health perspective (Hiono et al., 2022; Isoda et al., 2022; Rijks et al., 2021). With reports in wild birds, poultry, and mammals extending into Central and South America as of the fall 2022 (FAO, 2023), this H5N1 HPAI event has become panzootic and continues in 2023. Thus far, all available sequences from Central and South American countries have been identified as genotype B3.2 with only one genotype B1.2 to date (data not shown).

5. Conclusions

The 2021/2023 H5N1 clade 2.3.4.4b HPAIV event has surpassed the 2014/2015 H5 clade 2.3.4.4c outbreak as the largest animal health emergency in U.S. history, involving 175 wild bird species (USDA-APHIS, 2023c), several virus reassortments, with spillover into poultry as well as mammals. Phylogenetic analyses have shown that the transmission of these viruses into poultry have largely been point sources from wild birds with limited farm-to-farm spread. As with previous outbreaks of Gs/GD H5 viruses, migratory wild birds continue to play a major role in the maintenance and spread of these viruses on an intercontinental scale. This study highlights the distinct epidemiology of the current outbreak and has implications for its control and management, emphasizing the importance of surveillance in wild birds in the efforts to prevent further incursions into poultry. The genotyping tool from this work may facilitate ongoing investigations as response continues. Further studies evaluating the fitness of the viruses from different H5 clade 2.3.4.4b genotypes in various bird species may help to better understand the ecology of this continually evolving virus lineage.

Funding

This research was supported by the U.S. Department of Agriculture (USDA), Agricultural Research Service (ARS) Project 6612-32000-081-00D and by the USDA, Animal and Plant Health Inspection Service (APHIS) Interagency Agreement #60-6040-6-005. This research was also supported in part by an appointment to the ARS Research Participation Program administered by the Oak Ridge Institute for Science and Education (ORISE) through an interagency agreement between the U.S. Department of Energy (DOE) and USDA, and by Chungbuk National University BK21 program (2023). ORISE is managed by ORAU under DOE contract number DE-SC0014664. The findings and conclusions in this publication are those of the authors and the U.S. Geological Survey and do not be necessarily represent the official policy of the USDA, DOE, ORAU/ORISE. Any use of trade, product or firm names is for descriptive purposes and does not imply endorsement by the U.S. Government. The USDA is an equal opportunity provider and employer.

Data availability

Sequences generated by this study have been deposited in GenBank under accession numbers OQ957561 to OQ968080; raw sequencing data files will be available at the Sequence Read Archive (SRA) https://www.ncbi.nlm.nih.gov/sra.

CRediT authorship contribution statement

Sungsu Youk: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Mia Kim Torchetti:** Conceptualization, Methodology, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Kristina Lantz:** Methodology, Investigation, Validation, Formal analysis, Data curation, Writing – review & editing. **Julianna B. Lenoch:** Resources, Writing – review & editing. **Mary Lea Killian:** Methodology, Investigation, Validation, Data curation, Writing – review & editing. **Christina Leyson:** Validation, Data curation, Writing – review & editing. **Sarah N. Bevins:** Resources, Writing – review & editing. **Krista Dilione:** Resources, Writing – review & editing. **Hon S. Ip:** Resources, Writing – review & editing. **David E. Stallknecht:** Resources, Writing – review & editing. **Rebecca L. Poulson:** Resources, Writing – review & editing. **David L. Suarez:** Formal analysis, Funding acquisition, Writing – review & editing. **David E. Swayne:** Funding acquisition, Writing – review & editing. **Mary J. Pantin-Jackwood:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully acknowledge the tireless efforts of so many who have contributed to responding to this event from the local to national levels including Wildlife Services biologists and specialists, responders, laboratorians and the National Animal Health Laboratory Network, Dawn Williams-Copland and Edna Espinoza (SEPRL), and Tod Stuber (NVSL computational biologist). We also thank Dr Yohannes Berhane from the National Centre for Foreign Animal Disease, Canadian Food Inspection Agency, for providing virus sequences used in our analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.virol.2023.109860.

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